

SOLIDWORKS DESIGN PORTFOLIO

SAHIL GUPTA

Mechanical Engineering Student / CAD Designer

Email: sahilgupta2004@gmail.com

Phone: +91 9625036962

PROFILE & SKILLS

About Me

- Mechanical Engineering student with hands-on experience in 3D modelling and CAD design using SolidWorks. Skilled in creating mechanical parts, assemblies, and engineering drawings. Interested in product design, manufacturing, and mechanical design engineering.

Technical Skills

- SolidWorks
- Part Modeling
- Assembly Design
- 3D CAD Modeling
- Excel , MS word, Power point
- Data cleaning with the help of Python tools
- python tools(pandas ,numpy, matplotlib)

Crankshaft Design

Designed a 3D model of a crankshaft using SOLIDWORKS.

Created the model based on standard engine component dimensions.

Features Used

Extrude Boss/Base

Revolve Boss/Base

Fillet

Chamfer

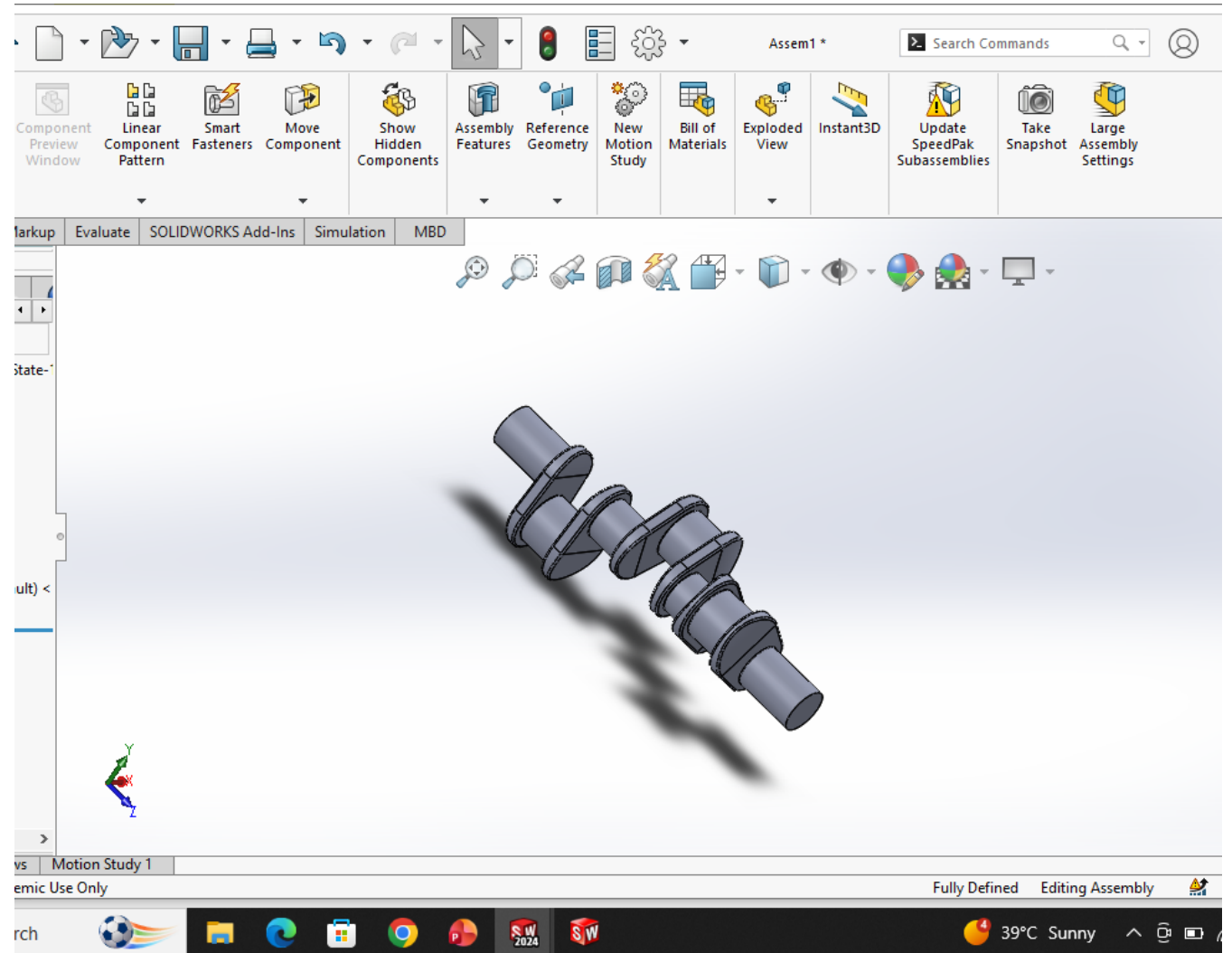
Mirror

Reference Geometry

Skills Demonstrated

3D Part Modeling

Mechanical Component Design



A crankshaft is a critical engine component that converts the up-and-down (reciprocating) motion of pistons into rotational (circular) motion.

Camshaft Design

Overview

Model a camshaft with multiple cam lobes in SOLIDWORKS.

Designed shaft and lobe profiles according to mechanical design requirements.

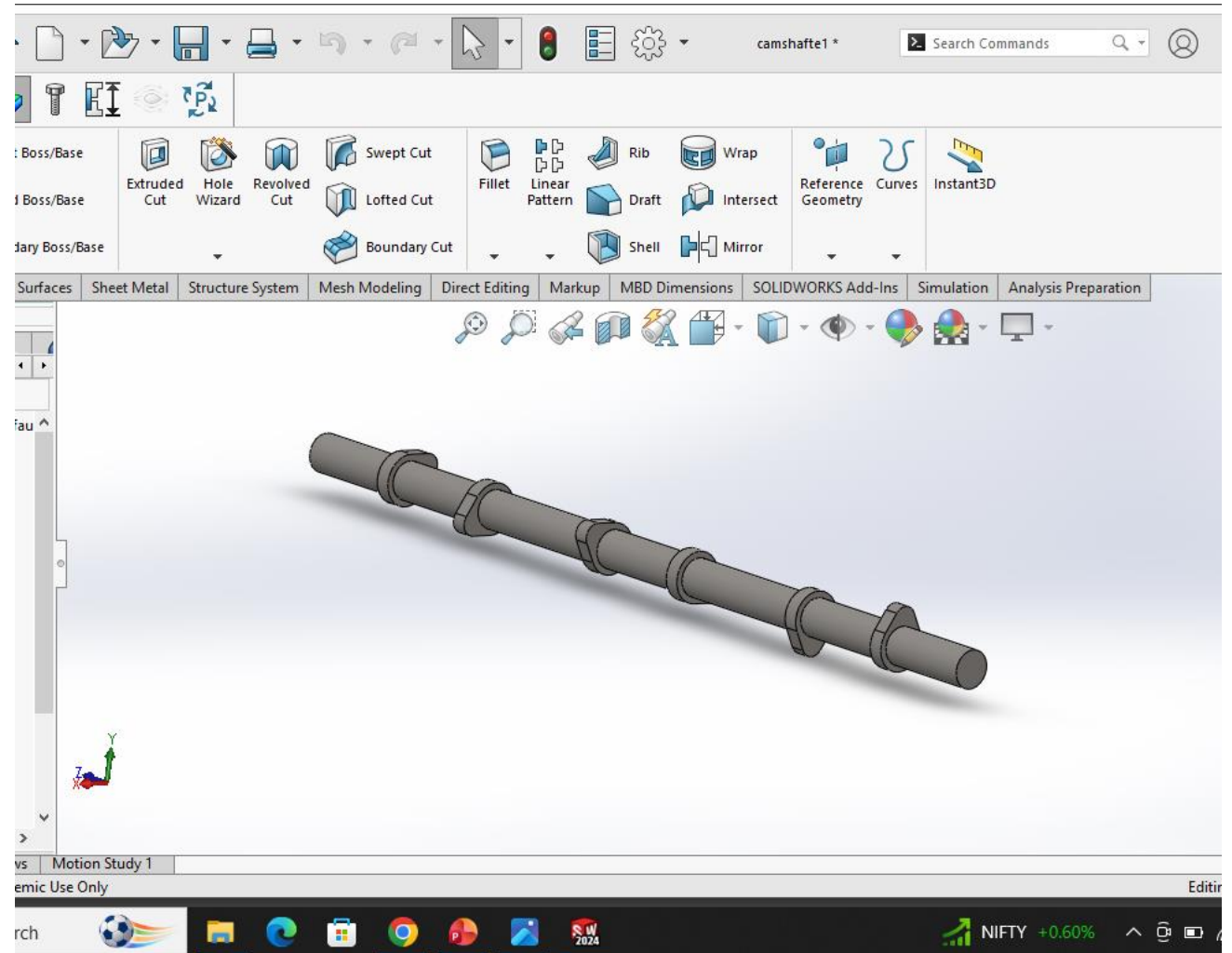
A camshaft is an engine component featuring egg-shaped bumps called "lobes". Its primary use is to act as a mechanical timekeeper that precisely controls the opening and closing of an engine's intake and exhaust valves

Features Used

Extrude Boss/Base

Revolve Boss/Base

Circular Pattern



Ball Bearing Design

Overview

Designed a 3D model of a standard ball bearing in SOLIDWORKS.

Model inner race, outer race, balls, and cage with proper assembly constraints.

Components Model

Inner Ring (Inner Race)

Outer Ring (Outer Race)

Balls (Rolling Elements)

Cage / Retainer

Features Used

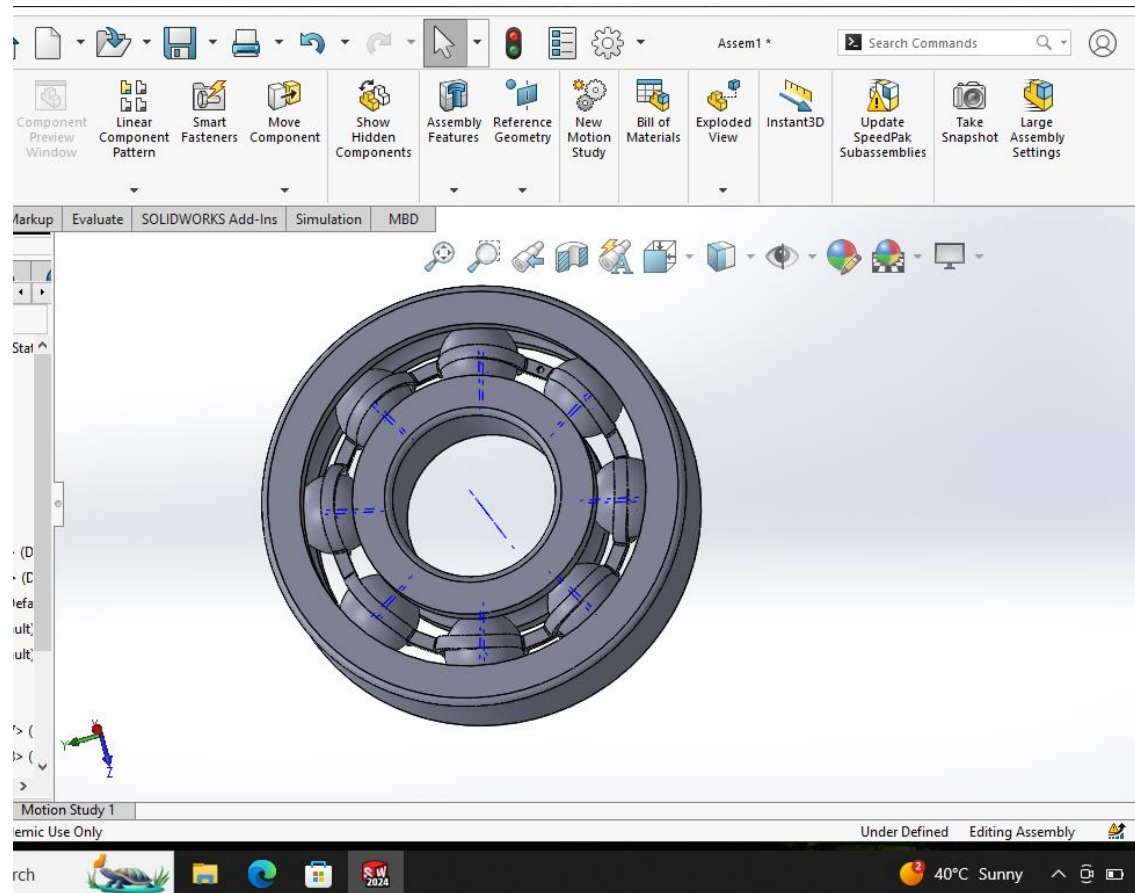
Revolve Boss/Base (for races)

Circular Pattern (for balls)

Assembly Mates (Concentric, Coincident)

Mirror (if applicable)

Reference Geometry



Ball bearings are used to **reduce rotational friction** and **support both radial and axial loads** in machinery. By converting sliding friction into rolling friction, they minimize wear and tear, prevent energy loss, and allow moving parts to rotate smoothly and efficiently.

Muff Coupling Design

Overview

Designed a 3D model of a Muff (Sleeve) Coupling in SOLIDWORKS.

Created a solid sleeve type coupling used to connect two co-axial shafts.

Ensured proper fit and alignment for torque transmission.

Components Model

Left Shaft

Right Shaft

Muff / Sleeve (single cylindrical body)

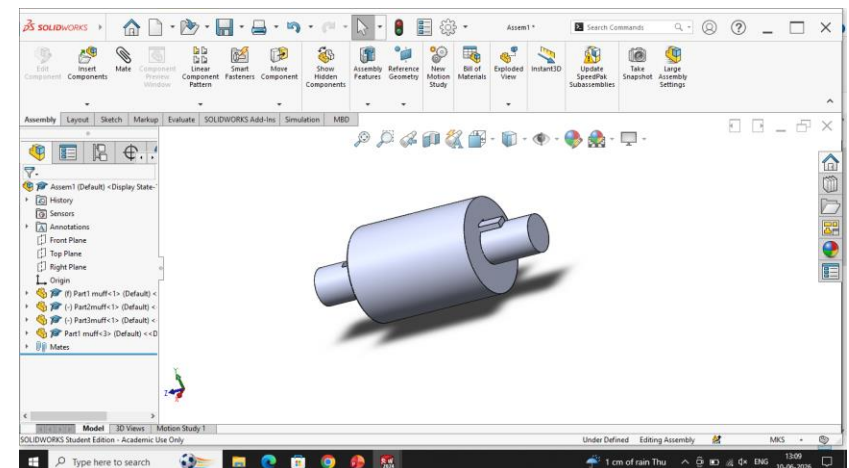
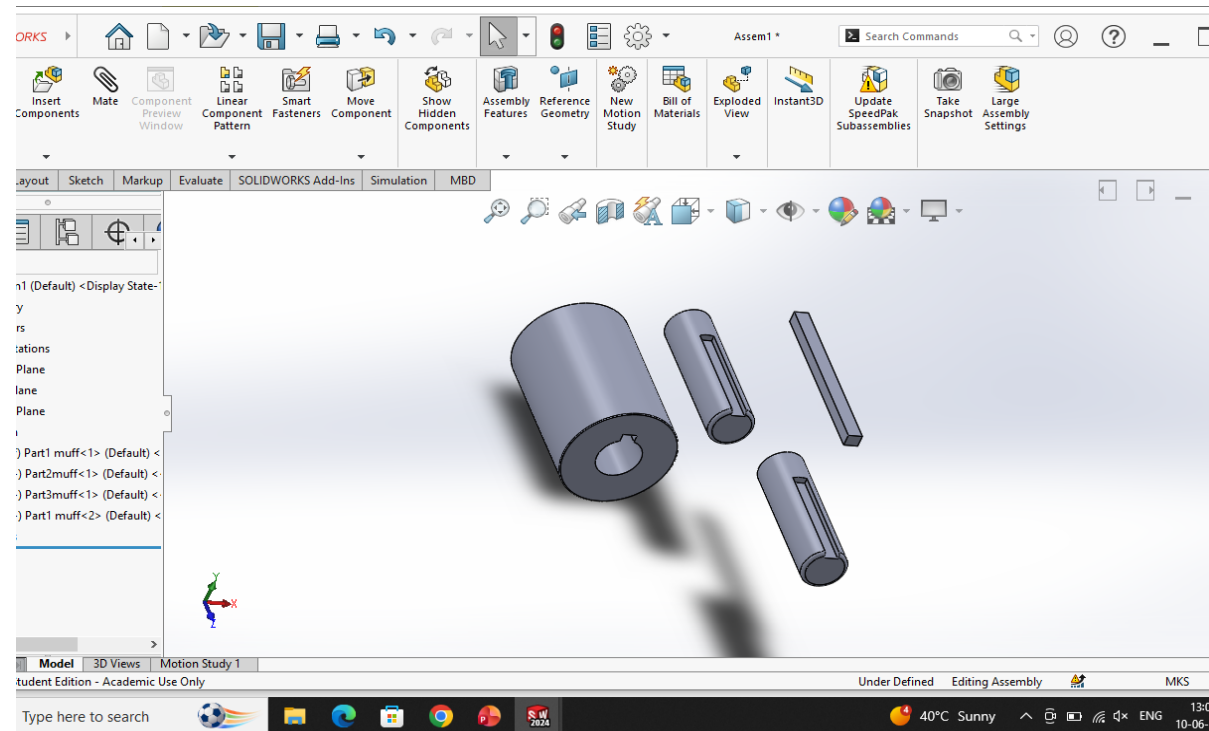
Features Used

Extrude Boss/Base (shaft and sleeve)

Hole/Extruded Cut (if keyway is added)

Fillet (for edge finishing)

Mirror (if symmetrical features used)



Conclusion

- Completed multiple mechanical components using SOLIDWORKS including crankshaft, camshaft, ball bearing, and muff coupling.
- Gained strong understanding of 3D part modeling, assembly design, and engineering drawings.
- Improved skills in precision modeling and mechanical design concepts.
- Able to apply CAD knowledge to real-world mechanical components.